

AI-103 Developing AI Apps and Agents on Azure.- Complete Certification Course

A production-style Azure AI engineering course focused on Foundry apps, agents, RAG, tools, multimodal AI, evaluations, monitoring, and information extraction.

Learn

Practice

Evaluate

Prepare

AI-103 Developing AI Apps and Agents on Azure - Complete Certification Course

A production-style Azure AI engineering course focused on Foundry apps, agents, RAG, tools, multimodal AI, evaluations, monitoring, and information extraction.

Best fit for developers and AI engineers who want to build agentic solutions with Microsoft Foundry and Azure AI services.

Product overview

This CloudXeus course prepares learners to design, build, evaluate, secure, and operationalize AI apps and agents on Azure. It uses Microsoft Foundry as the central platform and connects generative AI, RAG, tool use, multimodal AI, evaluation, observability, and Content Understanding into one practical learning path.

Product field	Value
Course code	AI-103
CloudXeus SKU	CX-AI103-APPS-AGENTS-2026
Course title	AI-103 Developing AI Apps and Agents on Azure - Complete Certification Course
Primary audience	Azure AI engineers, Python developers, cloud developers, solution builders, and technical architects building AI apps or agents
Level	Associate / intermediate to advanced
Format	Technical video lessons, Python demos, Foundry portal walkthroughs, agent labs, evaluations, and capstone project assets
Estimated learning time	24 hours video plus 14 guided labs and a capstone project
Access period	12 months from enrollment

Product field	Value
Instructor support	Q&A forum support with lab troubleshooting guidance
Practice assets	3 practice tests, project files, lab guides, sample prompts, and revision PDFs
Voucher included	No. This product does not include a Microsoft exam voucher.

Who this course is for

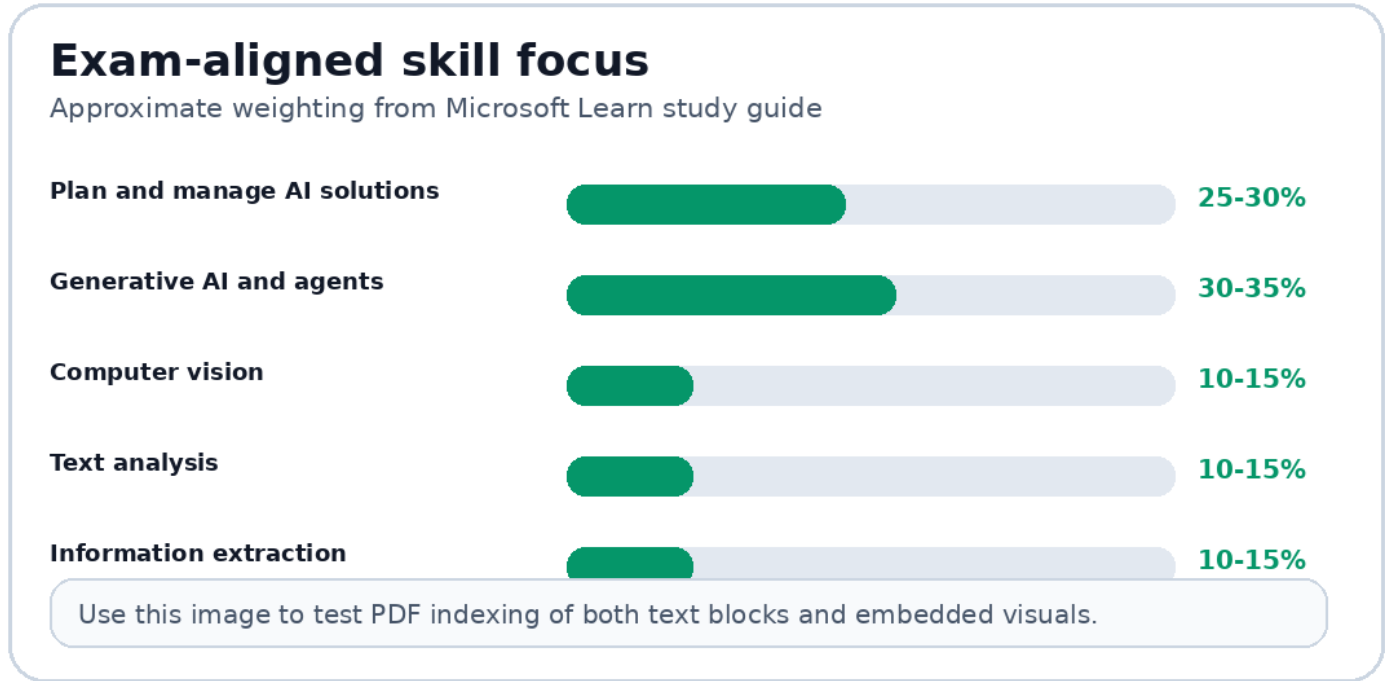
- Developers building AI apps and agents with Azure and Microsoft Foundry.
- Students who already understand basic AI concepts and want production-grade implementation skills.
- Cloud professionals who need RAG, evaluations, observability, safety, and agent tooling experience.

Prerequisites

- Experience developing applications with Python.
- Familiarity with general AI, generative AI, and core Azure services.
- An Azure subscription and Microsoft Foundry access are recommended for labs.

Exam alignment and course coverage

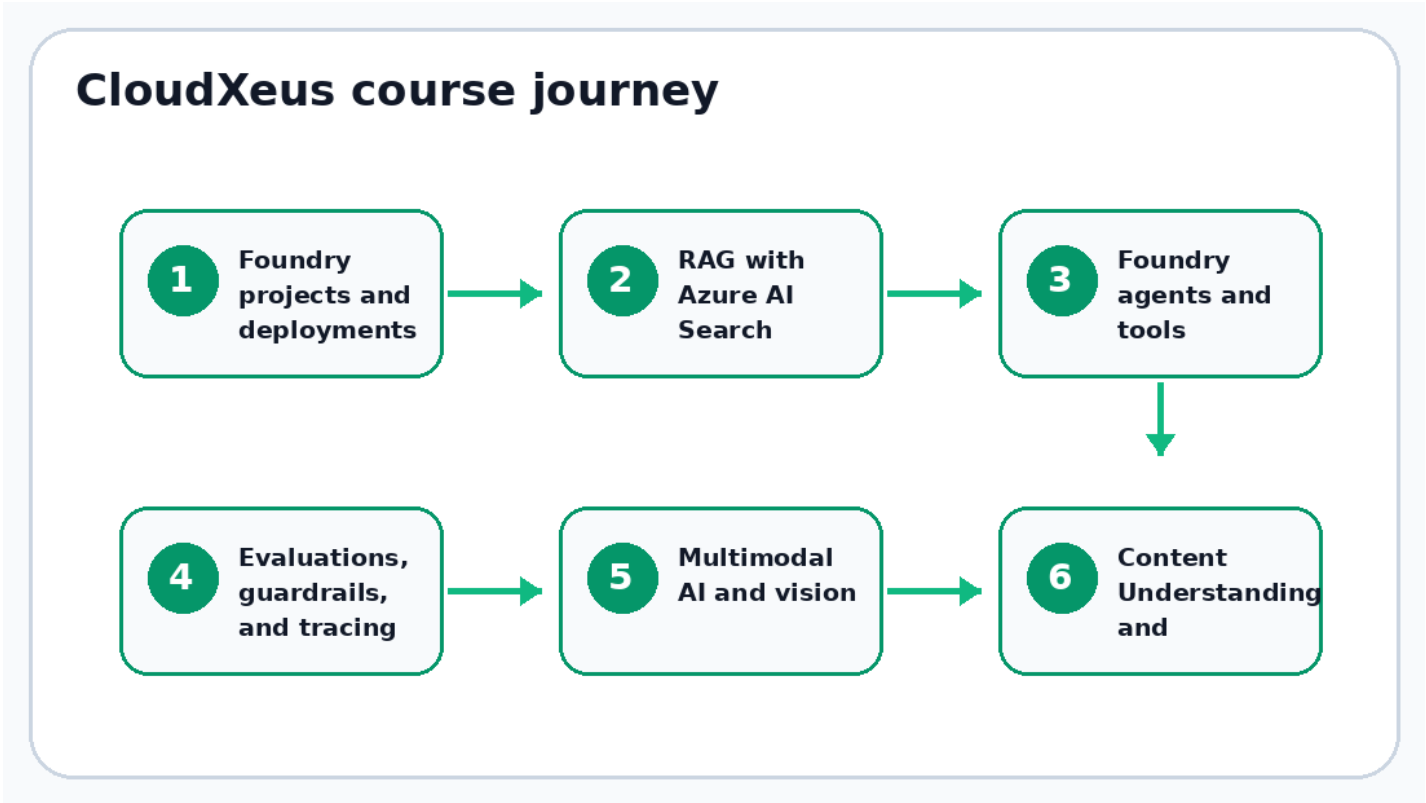
Aligned to the official Microsoft Learn study guide for AI-103, with skills measured as of April 16, 2026.



Skill area	Official weighting	How the course prepares students
Plan and manage an Azure AI solution	25-30%	Covers Foundry service selection, models, deployments, grounding, retrieval, indexing, memory, tool integration, quotas, monitoring, security, evaluators, guardrails, and governance.

Skill area	Official weighting	How the course prepares students
Implement generative AI and agentic solutions	30-35%	Builds RAG apps, Foundry apps, single-agent and multi-agent workflows, tool-augmented agents, memory-enabled agents, and agent monitoring patterns.
Implement computer vision solutions	10-15%	Explains multimodal prompts, image understanding, image generation, and vision-enabled app scenarios.
Implement text analysis solutions	10-15%	Covers text analysis scenarios such as key phrase extraction, entity recognition, sentiment, language detection, and integration patterns.
Implement information extraction solutions	10-15%	Uses Content Understanding and document intelligence style scenarios to extract structured information from documents, forms, images, audio, and video.

Module map



Detailed course content

Module	What students learn
Module 1: Foundry projects and deployments	Create Foundry projects, choose and deploy models, understand model deployment types, connect applications to projects, and manage configuration securely.
Module 2: RAG with Azure AI Search	Build retrieval-augmented generation solutions using Blob Storage, Azure AI Search, vector search, hybrid retrieval, semantic ranking, and grounded answers.
Module 3: Foundry agents and tools	Create agents, define roles and goals, integrate tools, implement function calling, connect knowledge stores, and design multi-agent workflows.
Module 4: Evaluations, guardrails, and tracing	Run quality and RAG evaluations, configure guardrails, inspect traces, use Application Insights, analyze failures, and improve prompts and retrieval behavior.
Module 5: Multimodal AI and vision	Use multimodal models for image input, image generation concepts, speech inputs, and real-world cross-modal scenarios.
Module 6: Content Understanding and information extraction	Use Azure Content Understanding to extract structured fields from documents, invoices, images, audio, and video for downstream agent workflows.

Hands-on labs and assets

Lab asset	Description
Lab 1 - Create a Foundry project and deploy a model	Configure a project, choose a model deployment, and connect from Python.
Lab 2 - Build RAG from Blob Storage and AI Search	Index documents from Blob Storage and retrieve content for grounded model responses.
Lab 3 - Build a course support agent	Create an agent that answers using product sheets and refuses out-of-domain prompts.
Lab 4 - Add tools and function calling	Connect an agent to custom tools and observe tool invocation behavior.
Lab 5 - Evaluate groundedness and relevance	Create evaluation prompts and measure response quality against source documents.
Lab 6 - Trace and monitor agent runs	Connect traces to Application Insights and inspect latency, errors, and retrieval flow.
Lab 7 - Extract invoice fields	Use Content Understanding to analyze invoice PDFs and return structured fields.

Learning outcomes

- Design and build a Microsoft Foundry based AI app or agent that uses grounded retrieval.
- Connect agents to tools, knowledge, memory, and external applications.
- Evaluate quality, safety, groundedness, and relevance using repeatable test prompts.
- Monitor deployed agents, handle throttling, and troubleshoot retrieval or response issues.
- Extract structured information from documents and invoices for downstream workflows.

Support and product policies

Policy topic	CloudXeus product statement
Enrollment access	Learners receive access to course videos, downloadable lab guides, sample code or datasets where applicable, and revision notes for 12 months.
Instructor Q&A	Students can ask course-related questions about concepts, labs, setup steps, and troubleshooting. Response times vary by volume.
Exam updates	Course content is periodically reviewed against Microsoft Learn study guides and major service updates.
Practice tests	Practice questions are included for revision. They are not official Microsoft exam questions and should be used for preparation only.
Certificate	A CloudXeus completion certificate is available after course completion. It is not a Microsoft certification.
Refunds	Refunds follow the purchasing platform policy. If bought through Udemy, Udemy refund rules apply.
Exam voucher	No certification exam voucher is included unless explicitly stated in a separate promotion.
